

Assistive Smart Stick: Safe and Independent Mobility for the Visually Disabled Using Sensor Based Technology

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Abstract— To provide assistance to the visually impaired, we propose this paper which shows the creation and deployment of an intelligent walking stick which integrates a number of technologies to help such people in their everyday lives. The system uses ultrasonic sensors and a servo to aid in obstacle detection and an ESP32-CAM to help recognize the object in front of the user. The servo is used in a sweeping motion to achieve a broader perspective of the user's surroundings using an ultrasonic sensor and the second ultrasonic sensor is used to check for depths for user's safety. The image processing is done in the cloud using artificial intelligence and the feedback is provided via buzzer and speaker. The FreeRTOS feature of ESP32 has been used to facilitate in multitasking without causing interference to individual machinery and the lightweight and cost-effective design of the proposed stick makes it viable for practical usage.

Keywords—ultrasonic sensors, ESP32-CAM, cloud-based image analysis, real-time obstacle detection, FreeRTOS task management.

I. INTRODUCTION

Eyesight is among the most crucial components of human physiology. The key to our environment is through our eyes. A World Health Organization (WHO) research estimates that 285 million people worldwide are visually impaired, with 39 million of them being blind [1]. In accordance to the ICD-10, visual acuity of less than 3/60 (0.05) or equivalent visual field loss in the better eye with the greatest correction is considered blindness. This is equivalent to having impaired walk-about eyesight. Low vision is described as having visual acuity of less than 6/18 (0.3) but equal to or better than 3/60 (0.05) in the better eye with the greatest possible correction (visual impairment classes 1 and 2 in ICD-10) [2]. The International Classification of Diseases -

10 (ICD-10) used by the World Health Organization now defines blindness differently than the National Program for Control of Blindness (NPCB) in India. The NPCB definition of blindness is 20/200, however the ICD-10 classification with the 2006 revision uses 20/400 as the VA requirement. These two definitions differ in what qualifies someone as blind.[3]. There are three acknowledged forms of blindness: being unable to discriminate between two or more colors, a condition known as color blindness, which is characterized by an inability to distinguish between two or more colors; flash blindness, which is brought on by a sudden exposure to bright light; and inattentional blindness, which is characterized by an inability to perceive a salient object or feature due to lack of attention [5]. Cataracts, trachoma, and glaucoma are among the leading causes of blindness, accounting for almost 70% of blindness worldwide. Other causes of blindness include age-related maculopathy (ARM), diabetes mellitus, ocular traumas, and onchocerciasis (river blindness) [4].

The Electronic Travel Aid (ETA) is a device that emits signals to detect the surroundings within a predetermined range or distance. It provides the user with details about the nearby environment [6]. Wearable technology, smart canes, and portable gadgets are the many types of ETAs now in use. To recognize objects in the immediate environment, ETAs like NavGuide [11], K-Sonar [12], NavCane [7], Ultracane[13], GuideCane[10], and electronic mobility cane (EMC)[9] employ ultrasonic sensors. To improve devices, several technologies have been employed; the most essential ones are based on optical/vision systems and ultrasonic sensors [8]. According to a poll by Vidul V. Meshram et al., a white cane does not disclose information about the surroundings, according to 93.21% of respondents. A white

cane did not provide information regarding descending staircases, according to 42.11% of subjects. They claimed that because it is difficult to judge the steps with a white cane when descending, it was harder for them to descend than to ascend for them. Seventy-three percent of participants agreed that it is necessary to recognize colors and things [7].

In reaction to the flaws and weaknesses in the current Electronic Travel Aid systems, we suggest a smart stick designed specifically for those who are visually challenged which uses ultrasonic sensor for detecting high speed objects and stationary objects do not pose as much a threat as the objects approaching with very high speeds. The ultrasonic sensor is mounted on a servo motor which rotates from 0-180 degree of the stick user and we give the user audio input on the object's approach direction. Another ultrasonic sensor that is fastened to the stick's bottom end has been utilized which alerts the user about any depth, potholes and descending staircase. We have provided a buzzer alert for the depth sensor for the ease of the user to distinguish between the audio alerts. We have used a ESP32-CAM to capture the image of an object, identify the object and provide feedback via speaker.

II. LITERATURE REVIEW

We applied a multi-technological approach in relation to the complexities entailed in our project and heterogeneity in requirements. Technologically, they were subdivided into three; namely, object detection, object recognition, and feedback. This means more appropriate utilization of advantages gained from the different technologies and reduced downsides.

A. Object Detection

This project deals with ultrasonic sensors and a servo motor. N.Dey et al. [20] proposed a paper which presents creating and deploying a walking stick with an ultrasonic sensor for those with vision impairments, utilizing HC-SR04 module for obstacle detection and buzzer alerting, using PI microcontroller 16F877A. Ando B et al [21] and G. Prasanthi [30] proposed a paper where it is explained how a novel multisensor architecture might assist blind persons with activities related to urban transportation. A Smart Blind Stick was developed by Bhushan Chaudhari et al. [22] for blind individuals utilizing ultrasonic sensors, a camera, and an earphone/speaker to provide auditory feedback. In their work, Amit Kumar et al. [28] discuss the implementation of an ultrasonic blind walking stick controlled via Arduino and provides a vibrational response on demand. In most of the existing technologies different sensors are used to handle diverse tasks but in most of the existing systems they have attempted to detect obstacles in only a single direction most certainly, the front of the user. By utilizing a servo motor, we have extended the range of obstacle detection, from the extreme left to the extreme right providing more efficient obstacle detection and safety to the user.

B. Object Recognition

Using the ESP32-CAM module, we have tried to achieve object recognition. Cloud computing and image processing techniques have been applied for the task of object identification. Ali Mirarab et al. [24] and T. Bednarzet al.

[25] have discussed how cloud computing could benefit medical image processing, from lower cost, and scalability to performance and fault tolerance, including tools and techniques that enable these resources. Mokhled Altarawneh et al. [26] evaluated cloud platform environments for image processing algorithms, analysis, and storage capabilities with reference to architecture orientation and service level agreement. We have aimed to further reduce the cost of the system using artificial intelligence unlike most systems which handle image processing using specific algorithms running on the cloud. This can help to achieve even better image processing without the need to run high resource demanding algorithms in the cloud.

C. Feedback

The blind stick we propose makes use of a buzzer as well as a speaker module to give necessary auditory alerts to the user. A smart stick that employs an Arduino UNO and a voice module was presented by Mrs. R. Abiramee et al. [27] to assist those who are blind in navigating and identifying barriers, fire, and water. It also allows users to get real-time support. Voice module provides speech feedback. The usage of a buzzer is covered by Amit Kumar et al. [28] explains how their suggested Arduino and ultrasonic sensor-based blind stick uses a buzzer to give a warning signal. The frequency of the buzzer's beep varies according on the target's distance. The frequency of the beep buzzer increases with the size of the obstacle gap. A low-cost, lightweight device with a microcontroller that processes signals and uses beeping noises to warn visually impaired people of obstacles, water, and dark regions was proposed by E. J. Chukwunazo et al. [29]. The microcontroller receives, processes, and transmits the signals to the alarm system, which alerts the user to respond right away. T. Tirupal et al. make use of the Arduino UNO, ultrasonic sensor, infrared sensor, LCD display, voltage controller, and voice playback module [31]. Majority of the existing systems use either a buzzer, or a speaker or a voice module to give necessary alerts. The blind stick we propose gives the user aural input using a speaker module which provides the direction of approach of objects as well as provide the auditory clue about the object identified and a buzzer gives a beeping sound when depth is detected. All the auditory feedbacks are engineered in a way that they are easily differentiable from each other.

III. PROPOSED METHODOLOGY

A. Workflow

Fig. 1 depicts the system's process.

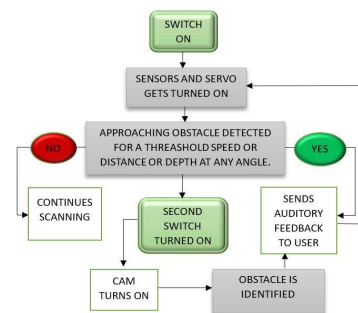


Fig. 1. Workflow

The stick has two switches one underneath the other. The user manually presses the switch which turns on the entire

system, the esp32 microcontroller board gets powered up from the 5V power source and the servo motor and ultrasonic sensors begin to operate. One ultrasonic sensor is mounted on the servo motor and the servo starts rotating from 0°-180° for the ultrasonic sensor to scan the environment for detecting high speed objects and impediments, the depth sensor simultaneously begins to search for potholes, descending staircases and depths. If any high-speed object is detected an auditory alert is sent to the user specifying the direction of approach of the object and a buzzer sounds if a depth is detected or an obstacle is found within 1 meter range. If no object or depth is detected the sensors continue scanning. If at any point the user feels the need to learn more about the object in front of them, he presses on the second switch where the second part of the system is activated the esp32 cam module is turned on and the Wi-fi of the user gets connected to the camera. The user receives the audio message specifying the contents of the image. It continues to take pictures till the camera is turned off. The user chooses whether to go in a certain route or halt after hearing the auditory feedback and notifications.

B. Block Diagram

The system's block diagram is seen in fig. 2.

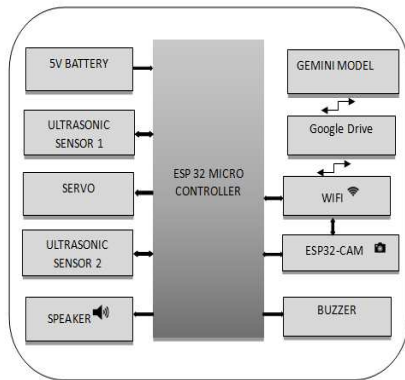


Fig. 2. Block Diagram

1) *Ultrasonic sensors for obstacle detection and speed calculation:* The time it takes for ultrasonic sound waves to bounce off of barriers is used off of barriers is used by the ultrasonic sensors to determine the distance. To activate the sensors, a pulse is sent out by the output pins known as Trig pins, the Echo pins are input pins which receive the reflected waves and we calculate the distance of the obstacle using the time taken for the waves to get reflected back using the formula:

$$\text{Distance} = \frac{\text{pulse time} \times \text{speed of sound}}{2}$$

Sound travels at a speed of 0.0343 cm/μs. The reason for the division by two is that we must determine the one-way distance since the waves reach the object and reflect back. Since the distance is measured in cm, the sound speed is measured in cm/μs. The maximum distance that ultrasonic sensors can record is 400 cm since their detection range is 2 cm to 400cm as per datasheet. If we put 400 cm as distance in the above formula, we find the maximum time the sensor should take to receive the reflected wave, which comes to approximately 23300 μs. This number is useful because we are utilizing a servo motor to trace the surroundings of the

user. 0 degrees to 45 degrees indicate the far left, 45 degrees to 75 degrees indicate left, 75 degrees to 105 degrees denote the front, 105 degrees to 135 degrees indicate right and 135 degrees to 180 degrees indicate the far right of the user. The servo sweeps the surrounding from 0 to 180 degrees with small increments of 30 degrees since the measuring angle of the sensor is about 30 degrees (at 0 degree position the sensor traces from -15 degrees to 15 degrees, at 30 degree position, the sensor traces from 15 degrees to 45 degrees and so on) and waits for a maximum of 23300 μs at every angle for the sensor to take the distance readings, beyond which we can conclude there is no object in the current direction in the given range and let the servo proceed to its next position. We have placed this servo at around knee level and mounted an ultrasonic sensor on the servo blade to check for all kinds of objects (refer to fig. 3). We are not concerned about the objects coming from behind, i.e., at angles beyond 195 degrees (at 180 degree position the sensor still traces from 175-195 degrees though it is not accurate above 187.5 degrees since the effectual angle of the sensor is about 15 degrees) to avoid too many cases and reduce the time it takes for the servo for a full 180 degrees sweep. If an object is in front of the user at a very close distance, we notify the user using a buzzer sound, if an object is present in static condition in any other direction, we ignore them as the user is not at risk of colliding with such objects, and if an object is approaching towards the user from any direction within 0-180 degrees greater than a high threshold speed, we send audio feedback specifying the direction. To calculate the speed at a certain angle, the sensor checks for a distance reading at that angle, and if the sensor obtains a distance value within 23300 μs, a next reading is taken at that same angle after a very small delay (100-500 ms). The formula for speed calculation is as follows:

$$\text{Speed} = \frac{\text{distance1} - \text{distance2}}{\text{time}}$$

where, $\text{time} = \frac{\text{delay}}{1000}$

Here, delay is in milliseconds.

The ultrasonic sensor fitted near the bottom of the stick facing the ground works independently and checks for distance values every second, and if a value greater than a threshold value (average distance from ground level with considerations of stick angle) is found, we notify the user with a buzzer sound of different frequency (to distinguish from obstacles in front). This threshold value can be modified using the right potentiometer. The two potentiometers are connected in a voltage divider circuit, so the two potentiometers do not work independently. So, we fix the left potentiometer knob at the lowest and change the threshold by using the right potentiometer knob.

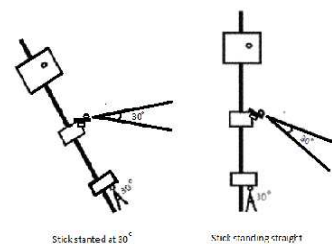


Fig. 3. Different stick positions

Both the tasks run independently using the FreeRTOS feature of the ESP32 and do not interfere in the operation of one another. We give no auditory feedback for 5 seconds upon receiving any feedback to allow the user to perceive their surroundings (too many feedbacks at once might confuse the user).

2) *Esp32 CAM module for object identification*: The OV2640 is an onboard camera on the ESP32 CAM development board. The camera has a 78° field of view [14] as shown in fig. 4. The user must hold the stick straight in order for the camera to take a photo in front of them. According to the specification sheet, the camera's acquired picture is in JPEG, BMP, and grayscale formats. We chose a powerful flash with a 310-mA power consumption at 5V to improve picture clarity, but we want to optimize it for lower power consumption soon.

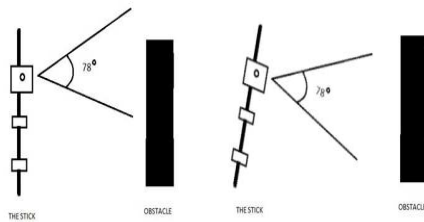


Fig. 4. Viewing angle of the OV2640 camera

Utilizing the Arduino IDE, we have developed an algorithm that links the ESP-32 CAM to Google Drive through a web application. The method is made for the ESP-32 CAM module, which utilizes a Google Apps Script deployment to transmit a picture to Google Drive when the button for camera is pressed. The Wi-Fi is accessed by the ESP-32 with a password and SSID. Once the GPIO pins and parameters are setup, the camera module is initialized. Once the upload to Google Drive has been confirmed, the CAM module determines if it can connect to Google Apps Script. ESP-32 takes a picture which is base64-encoded and then the drive requests the image via HTTP from Google Apps Script. The Google drive is mounted in the Google Colab environment to access data stored in the directory. The code that runs on the Google Colab environment looks for the specified folder for picture files and finds the most current one. Using a Gemini API request, the detected picture is submitted to the Gemini model for analysis. After then, the Gemini model explains the image's content in accordance with the given prompt. Following the successful picture classification and identification process, we provide the user with audio feedback regarding the image's contents.

C. Circuit Diagram

The full system's operational circuitry is seen in fig. 5.

1) *Microcontroller ESP32 DevKit V1*: The ESP32 DevKit V1 is chosen based on several key features which include dual core processor supporting multiple tasks using FreeRTOS, Wi-Fi and Bluetooth connectivity, multipurpose GPIO pins and low cost. The ESP32, which serves as the primary microcontroller, a 5V DC power source is used and programmed using Arduino IDE to achieve the functionalities of the smart blind stick.

2) *HC-SR04 ultrasonic sensor*: When the TRIG pin is set to HIGH for $10\mu s$, it delivers an eight cycle sonic burst at 40 kHz as per the HC-SR04 datasheet. Until an ultrasonic burst is detected again, the ECHO pin remains HIGH; after then, it will drop to LOW. Any GPIO pin of the ESP32 may be linked to the TRIG and ECHO pins of the HC-SR04. The GND pin must be linked to the common ground, while the VCC pin is linked to a +5V supply [14][15].

3) *SG90 Micro Servo Motor*: The SG90 Micro Servo which can rotate approximately 180 degrees (90 degrees in each direction) as per its datasheet is positioned by use of Pulse Width Modulation (PWM) signals which determines the angle to which the servo rotates. The red VCC wire of the servo is linked to the +5V supply and the brown GND wire is linked to the common ground. The servo's orange signal wire can be connected to any GPIO pin of the ESP32 which can generate PWM signals, D33 in our case [16].

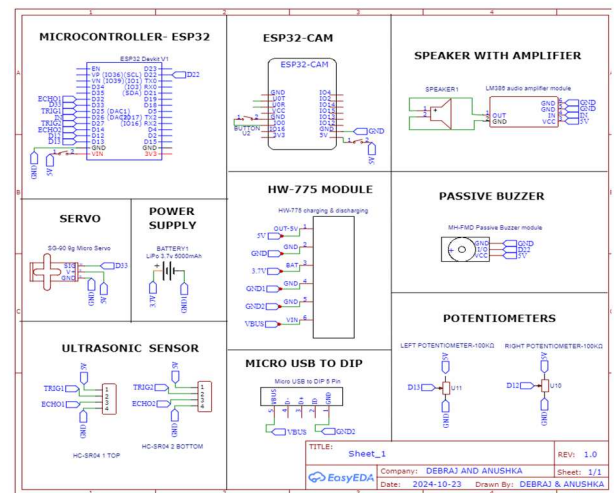


Fig. 5. Circuit Diagram

4) *ESP32-CAM*: The ESP32-CAM is a compact microcontroller integrating an ESP32-S microcontroller with an OV2640 2 MP camera module. The built-in Wi-Fi and bluetooth connectivity makes it popular in IoT and AI-based vision applications. The 5V or VCC pin of the ESP32-CAM is linked to the +5V supply and its GND pin is linked to common ground to integrate it into the system.

5) *Speaker with LM386 audio amplifier module*: An LM386 module is connected to a tiny speaker module [17]. The amplifier module's two GND pins are linked to the common ground, the VCC pin is linked to the +5V supply and its IN pin is linked to a GPIO pin on the ESP32 capable of generating analog output (via Digital-to-Analog Converter, DAC), pin 26 in our case.

6) *MH-FMD Passive Buzzer module*: The I/O pin of the buzzer module can be connected to any GPIO pin of the ESP32 capable of generating a PWM signal. To incorporate the buzzer, the GND pin is linked to common ground and the VCC pin is linked in the +5V power source [32].

7) *Potentiometers*: Two 100k ohm potentiometers are provided to manage the threshold depth value required for our smart blind stick. The left potentiometer knob is fixed at the lowest and the right potentiometer knob is used to control the ultrasonic sensor's threshold value at the stick's base. The middle pins of the left and right potentiometers

are connected to the GPIO 13 and 12 of the ESP32 respectively.

8) *5V rechargeable power supply*: A 3.7V Li-ion battery is connected to a HW-775 charging & discharging module which helps to both output 5V for our system power supply as well as charge the battery at 5V. This is achieved by using a Micro USB to DIP 5 pin connector which comes with a D type port to charge the battery using a 5V 2A adapter.

IV. RESULTS AND DISCUSSIONS

Fig. 6 shows the assistive stick which stands 110cm tall with the top ultrasonic sensor at 65 cm from ground level and bottom ultrasonic sensor at 20 cm from ground level. The camera is fitted through a hole made in the box and is at 70 cm from ground level. The speaker is fitted at the left side of the box. The microcontroller board, 5V power supply along with other components is inside the box.

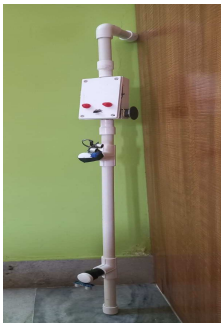


Fig. 6. Smart Blind Stick

The graphs in fig. 7 and fig. 8 indicate a set of readings from the two ultrasonic sensors. Fig. 7 shows the comparison of the simultaneous distance readings from the top and bottom sensors along with the points at which alert is triggered as shown by “x” mark in the graph.

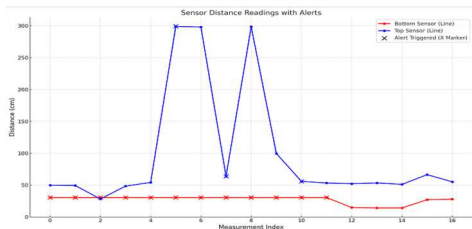


Fig. 7. Both Sensor Distance Readings with Alerts

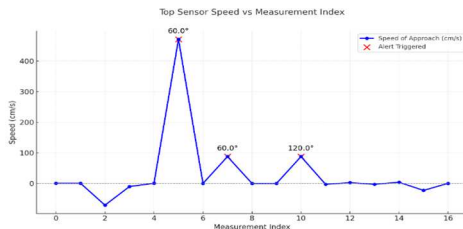


Fig. 8. Top Sensor Speed Readings with Alerts

Fig. 8 shows the speed readings of the top sensor at the same points from fig. 7, with “x” mark at points of high-speed showing alerts triggered along with the sensor position (angle) at those points. The graphs have been plotted without providing any system delay to test the simultaneous working of the top and bottom sensors. The final system issues suitable delay between alerts.

Fig. 9 shows a blue water bottle and fig. 10 shows a black colored backpack. These pictures were captured by ESP32-CAM which was then sent to Google Drive and fetched by the image analysis algorithm.



Fig. 9. Image captured by cam

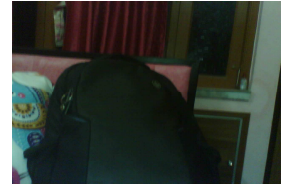


Fig. 10. Image captured by cam

In fig. 11 and fig. 12, we can see the output generated by the image analysis model where it has correctly analyzed the contents of the image.

```
send: b'GET /v1beta/files/jt7ip04nb70?%24alt=json%3Benu
reply: 'HTTP/1.1 200 OK\r\n'
header: Server: TornadoServer/6.3.3
header: Content-Type: text/html; charset=UTF-8
header: Date: Mon, 21 Oct 2024 16:09:05 GMT
header: Etag: "8ee6e2292e5bc63b426c8f5e35557d7fe642496"
header: Content-Length: 450
Uploaded file 'waterbottle.jpg' as: https://generativela
send: b'POST /v1beta/models/gemini-1.5-pro-002:generateC
send: b'{"contents": [{"parts": [{"text": "The closest item in the image is a blue water bottle."}]}
reply: 'HTTP/1.1 200 OK\r\n'
header: Server: TornadoServer/6.3.3
header: Content-Type: text/html; charset=UTF-8
header: Date: Mon, 21 Oct 2024 16:09:08 GMT
header: Content-Length: 281
```

Fig. 11. Output produced for image in fig. 9

```
header: Server: TornadoServer/6.3.3
header: Content-Type: text/html; charset=UTF-8
header: Date: Mon, 21 Oct 2024 15:55:38 GMT
header: Etag: "dc93f93e1c38d2d674a6e7187ff1559862c1d669"
header: Content-Length: 445
Uploaded file 'object.jpg' as: https://generativela
send: b'POST /v1beta/models/gemini-1.5-pro-002:generate
send: b'{"contents": [{"parts": [{"text": "The closest item in the image is a black backpack."}]}
reply: 'HTTP/1.1 200 OK\r\n'
header: Server: TornadoServer/6.3.3
header: Content-Type: text/html; charset=UTF-8
header: Date: Mon, 21 Oct 2024 15:55:41 GMT
header: Content-Length: 263
```

Fig. 12. Output produced for image in fig. 10

This output will then be told to the user via the speaker.

V. CONCLUSION

The smart blind stick project shall ease and ensure navigation for the visually impaired in their daily avocations-a humanitarian pursuit. Standing for real-time obstacle detection and phase-out auditory feedback, very simple and effective technology could stand out as the combination of ultrasonic sensors with a depth sensor, servo motors, and ESP32-CAM. This version of the device will allow it to detect some obstacles, indicate dangers, and provide auditory cue about the object in front of the user via the camera module. However, it can only be termed as an ongoing project. Also, the future could lead to more developments in object detection, getting feedback, which in turn will further refine the device thereby making it user-friendly and also very trustworthy.

VI. FUTURE SCOPE

- Without requiring mechanical movement, the electrically scanned array would provide quick, real-time, and extremely accurate obstacle recognition across a large range of angles.
- It increases the speed and accuracy of detection by lowering mechanical wear and tear [18].
- Isolating individual potentiometers using optocouplers would improve electrical isolation and noise

immunity between different system components. By providing such isolation, cross-talk between the signals is prevented [19].

- We can enhance the design of the stick in such a way that the stick can become a foldable one.
- To precisely identify fast-moving objects, a Doppler radar module might be included into the system.
- For operation of the stick in noisy environments, we can use bluetooth headset and earphones instead of speaker.
- Power consumption of the system may be further optimized.

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